CHAPTER 5

164 General comment about Chapter 5: The historical aspects of this chapter ate based in large part (i) on my interviews with participants in the events described, or with their scientist colleagues and friends (INT-Baym, INT- Braginsky, INT-Eggen, INT-Fowler, INT-Ginzburg, INT-Greenstein, INT-Harrison, INT-Khalatnikov, INT-Lifshitz, INT-Sandage, INT-Serber, INT-Volkoff, INT-Wheeler), and (ii) on my reading of the scientific papers the participants wrote. For general background on the history of physics in the 1920s and 1930s, I have relied somewhat on Kevles (1971), and for background on the history of Soviet physics, on Medvedev (1978). Useful information and background about Landau came from Livanova (1980) and Gamow (1970), about Oppenheimer from Rabi et al. (1969) and Smith and Weiner (1980), and about the development of Wheeler's ideas from his research notebooks, Wheeler (1988). In some places I have relied on other sources cited below.

164 ["By the time I knew Fritz . . . wrong,"] INT-Fowler.

164 [Jesse Greenstein . . . "a self-proclaimed genius . . . other people."] INT-Greenstein, and Greenstein (1982).

165 [He even went on the air . . . popularize his neutron stars.] Zwicky (1935).

166 ["Zwicky called Baade . . . same room," recalls Jesse Greenstein.] INT- Greenstein.

168 [(Today we know, . . . factor of 10,] Baade (1952).

168 [By combining Baade's knowledge . . . larger factor, 10 million,] These are Baade and Zwicky's numbers, as they appear in the abstract of a talk that is replicated in Figure 5.2 (Baade and Zwicky, 1934a), except for the "10,000 and perhaps 10 million," which come from their more detailed paper on the issue (Baade and Zwicky, 1934b). Their error resulted from assuming that when the supernova is brightest, the circumference of its hot, radiating gas is in the range of 1 to 100 solar circumferences. In fact, the circumference is far larger than this, and when one traces through their argument, this results in far less ultraviolet light and X-rays.

171 [The neutron arrived . . . it seemed to Zwicky.] In this section and throughout Chapter 5, I attribute to Zwicky the concept of a neutron star and its consequences for supernovae and cosmic rays, although the publication of the ideas was joint with Baade. Giving Zwicky the credit for the ideas (and Baade the credit for the key understanding of the observational data) is an informed speculation based on my discussions with their scientist colleagues: INT-Eggen, INT-Fowler, INT-Greenstein, INT-Sandage.

174 Figure 5.2: Baade and Zwicky (1934a). For some justification of the numbers in the abstract see the more detailed presentation in Baade and Zwicky (1934b).

178 [Landau's publication . . . cry for help:] This interpretation of Landau's publication was explained to me by his closest, lifelong friend, Evgeny Michailovich Lifshitz (INT-Lifshitz).

180 [A fellow postdoctoral. . . "I vividly remember . . . of the paper."] Quoted in Livanova (1980).

180–181 ["All the nice girls . . . are left,"] Quoted in Livanova (1980).

181 [As George Gamow, . . . "Russian science . . . capitalistic countries."] Gamow (1970).

181 [In 1936 Stalin, . . . were destroyed.] The statistics on imprisonments and deaths under Stalin are somewhat uncertain. Medvedev (1978) gives what are perhaps the most reliable numbers available in the 1970s. However, in the late 1980s glasnost made possible the public dissemination of information that drove the numbers upward. The